1

$$\begin{bmatrix} y_1' = 5y_1 - 6y_2 \\ y_2' = 2y_1 - 2y_2 \end{bmatrix} \begin{bmatrix} y_1(0) = 5 \\ y_2(0) = 3 \end{bmatrix}$$

$$A = \begin{pmatrix} 5 & -6 \\ 2 & -2 \end{pmatrix}$$

vl. čísla:

$$\det \begin{pmatrix} 5-\lambda & -6 \\ 2 & -2-\lambda \end{pmatrix} = \lambda^2 - 3\lambda + 2 = (\lambda - 2)(\lambda - 1)$$

$$(\lambda - 2)(\lambda - 1) = 0$$

$$\lambda = 1, 2$$

$\lambda = 1$:

$$\begin{pmatrix} 4 & -6 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \vec{0}$$

$$\begin{pmatrix} 4 & -6 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} v_1 \\ 2 \end{pmatrix} = \vec{0} \longrightarrow \vec{y}_a = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \cdot e^x$$

$$\begin{pmatrix} 4v_1 - 12 \\ 2v_1 - 6 \end{pmatrix} = \vec{0}$$

$$v_1 = 3$$

$\lambda = 2$:

$$\begin{pmatrix} 3 & -6 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \vec{0}$$

$$\begin{pmatrix} 3 & -6 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} v_1 \\ 1 \end{pmatrix} = \vec{0} \longrightarrow \vec{y}_b = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot e^{2x}$$

$$\begin{pmatrix} 3v_1 - 6 \\ 2v_1 - 4 \end{pmatrix} = \vec{0}$$

$$v_1 = 2$$

OŘ:

$$\vec{y} = a\begin{pmatrix}3\\2\end{pmatrix}e^{x} + b\begin{pmatrix}2\\1\end{pmatrix}e^{-2x} = \begin{pmatrix}3ae^{x} + 2be^{-2x}\\ 2ae^{x} + be^{-2x}\end{pmatrix}$$

$$\left[\begin{array}{l} y_1(x) = 3ae^{x} + 2be^{-2x}, \\ y_2(x) = 2ae^{x} + be^{-2x}, \quad x \in \mathbb{R} \end{array}\right.$$

Poč. podm.

$$\cancel{y_1(0) = 3ae^{x} + 2be^{-2x} = 5}$$
$$\cancel{y_2(0) = 2ae^{x}}$$

$$y_1(0) = 3a + 2b = 5$$
$$y_2(0) = 2a + b = 3 \rightarrow b = 3 - 2a$$

$$3a + 2(3-2a) = 5$$
$$3a + 6 - 4a = 5$$
$$-a + 6 = 5$$
$$-a = -1$$
$$a = 1 \rightarrow b = 3 - 2a$$
$$b = 3 - 2$$
$$b = 1$$

$$\boxed{\left[\begin{array}{l} y_1(x) = 3e^{x} + 2e^{-2x}, \\ y_2(x) = 2e^{x} + e^{-2x}, \quad x \in \mathbb{R} \end{array}\right.}$$

②

$$\begin{bmatrix} y_1' = q y_2, \\ y_2' = -q y_1 \end{bmatrix}$$

$$A = \begin{pmatrix} 0 & q \\ 0 & -q \end{pmatrix}$$

vl. čísla

$$\det\left(\begin{pmatrix} 0-\lambda & q \\ 0 & -q-\lambda \end{pmatrix}\right) = \lambda^2 + q\lambda = \lambda(\lambda+q)$$

$$\lambda(\lambda+q) = 0$$

$$\lambda = -q, 0$$

<u>$\lambda = 0$:</u>

$$\begin{pmatrix} 0 & q \\ 0 & -q \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \vec{0}$$

$$\rightarrow \vec{y_a} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} q v_2 \\ -q v_2 \end{pmatrix} = \vec{0}$$

$$v_2 = 0$$

<u>$\lambda = -q$</u>

$$\begin{pmatrix} q & q \\ 0 & 0 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \vec{0}$$

$$\begin{pmatrix} q v_1 + q v_2 \\ 0 \end{pmatrix} = \vec{0}$$

$$\rightarrow \vec{y_b} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-qx}$$

$$v_1 = -v_2$$

$$\vec{y} = a\begin{pmatrix} D \\ 0 \end{pmatrix} + b\begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-ax} = \begin{pmatrix} aD + be^{-ax} \\ -be^{-ax} \end{pmatrix}$$

$$\boxed{\left[\begin{array}{l} y_1(x) = aD + be^{-ax}, \\ y_2(x) = -be^{-ax}, \quad x \in \mathbb{R},\ D \in \mathbb{R} \end{array}\right]}$$